

The National Information Ecosystem

Module OS A01 teaching presentation

AI assisted statistical work with accountable human judgement

By the end learners can	Required professional evidence
Define a statistical ecosystem boundary	Decision, population, geography, time and exclusions
Map actors, assets and critical flows	Stable identifiers and source-traceable records
Translate user needs into fitness requirements	Acceptance evidence and named owners
Use AI for bounded statistical assistance	Prompt, raw output, verification and corrections
Decide whether AI improves the workflow	Quality-adjusted comparison and recommendation

Official statistics emerge from connected decisions and controls

Ecosystem element	Material contribution	Typical failure
Institutions and authority	Mandates, access and release decisions	Conflicting or assumed authority
People and roles	Statistical, domain, operational and user knowledge	Role conflation or missing ownership
Information assets	Evidence across surveys, records and other sources	Unknown provenance or coverage
Flows and controls	Transformation, validation and escalation	Silent processing and weak lineage
Users and feedback	Fitness requirements and revision needs	Products disconnected from decisions

The statistical ecosystem boundary

Boundary dimension	Question	Kintaba example
Decision	What action will use the result?	Earlier food-security planning
Population	Who or what should the statistic represent?	Agricultural holdings, with documented exclusions
Geography	Where must results support decisions?	Forty-eight districts
Time	Which reference period and deadline apply?	Current crop season, six weeks earlier
Product	What statistical output is required?	Crop area and production with uncertainty
Exclusions	What remains outside the analysis?	Unreviewed release and automated eligibility

Actors and multiple roles

Actor	Roles in the ecosystem	Evidence boundary
KNSO	Official producer, method authority, release authority	Statistics Act E01
Agriculture ministry	Data holder and crop subject specialist	Mandate E02
District extension services	Operational supplier and local interpreter	Reporting SOP E04
Satellite analysis unit	Technical supplier	Pilot memo E05
Food-security planners	User and feedback source	Calendar and consultation E07 E09

One organization may hold several roles. Each role requires separate evidence.

Authority must be evidenced

- Legal mandate and delegated powers
- Method approval and access rights
- Release, revision and escalation decisions

Accountability remains named

- Statistician owns statistical validity
- Subject specialist owns contextual validity
- Other authorities cover law, security and confidentiality

Prompt authorship does not confer professional authority

Evidence layer	Purpose	Control question
Governing instruments	Establish mandate and authority	Is the instrument current and applicable?
Methods and standards	Define concepts, classifications and estimation	Which version governs this product?
Operational records	Describe the process actually performed	Does practice match the documented process?
Data and metadata	Support measurement and interpretation	Are provenance and transformations complete?
User evidence	Define needs and consequences	Are requirements explicit and testable?

Information assets and provenance

Required field	Why it matters
Owner and lawful access	Defines responsibility and permitted use
Population and coverage	Shows what the asset can represent
Concepts and classifications	Supports consistent interpretation
Identifiers and time	Determines linkage and temporal alignment
Transformations and versions	Provides reproducible lineage
Quality signals and limitations	Supports fitness decisions
Verification question	Directs independent review

Surveys censuses and total error

SOURCE ASSESSMENT

Error domain	Diagnostic question	AI contribution
Coverage	Who is omitted, duplicated or out of scope?	Draft comparisons from approved documentation
Measurement	Do questions and instruments capture the concept?	Flag inconsistencies for specialist review
Nonresponse	Which units or items are missing, and why?	Generate stratified diagnostic code
Processing	Where can coding, editing or transformation fail?	Draft validation tests
Estimation	Do weights and variance methods support the product?	Check code structure, not method authority
Revision	How will new evidence change published results?	Draft traceable revision summaries

Administrative records

- Administrative purpose differs from statistical purpose
- Population rules follow program operations and eligibility
- Concepts and identifiers can change when systems or policies change
- Process signals can reveal breaks before numeric diagnostics do
- Statistical use requires documented access, transformation and quality controls

Required product Administrative data quality report

Alternative and research sources

SOURCE ASSESSMENT

Source type	Potential value	Primary assurance question
Geospatial imagery	Frequent spatial evidence	What measurement process links pixels to the official concept?
Sensors	High-frequency observations	How do calibration, missingness and drift affect inference?
Private platform data	Timely behavioral signals	Which populations and actions generate the data?
Research datasets	New methods and experimental evidence	Can findings transfer to the production population and process?

Representativeness and generalization require evidence, not novelty

Integrated sources and uncertainty

Integration issue	Failure mechanism	Required evidence
Linkage	False and missed matches	Domain-specific linkage evaluation
Duplication	Units counted more than once	Duplicate rules and residual rates
Temporal alignment	Sources refer to different periods	Reference-time reconciliation
Classification	Different definitions or coding systems	Concordance and version controls
Uncertainty propagation	Input error disappears from final reporting	Combined uncertainty assessment

Flows lineage and control points

Flow record field	Example from Kintaba
Input and source	Monthly district spreadsheets
Transformation	Local compilation and central ingestion
Output	Agriculture ministry database
Control	Schema, range and version checks
Failure mode	Different district validation rules
Consequence	Comparability failure
Fallback and owner	Quarantine, district query and named processing lead

Semantic interoperability

- Shared concepts and definitions
- Classification concordances and versions
- Clear units, reference periods and populations

Technical interoperability

- Stable schemas and identifiers
- Machine-readable validation rules
- Versioned interfaces and reproducible transformations

Coherence requires both forms

User needs and fitness for use

QUALITY

Dimension	Example requirement	Acceptance evidence
Relevance	District crop results match planning decisions	Agreed concepts, crops and users
Accuracy	Error and bias acceptable by domain	Predefined validation and uncertainty criteria
Timeliness	Results arrive six weeks earlier	Deadline met after review gates
Coherence	Differences from survey series are explained	Reconciliation and revision bridge
Comparability	District and seasonal meanings remain stable	Domain tests and metadata
Accessibility	Authorized users can use the product safely	Disclosure and access controls
Interpretability	Users understand uncertainty and revisions	User-tested metadata and flags

Risk question	Resilience response
What single dependency can halt production?	Document fallback sources, methods and owners
Which skills or vendors create concentration risk?	Maintain internal competence and exit options
How will changes be detected?	Monitor process signals, versions and domain diagnostics
What happens when quality falls?	Apply stopping, quarantine and escalation rules
Can the institution sustain verification effort?	Measure total accepted effort, not demonstration speed

Confidentiality security and lawful use

- Classify data before selecting an AI tool or environment
- Exclude personal data, confidential unit records and credentials from unapproved services
- Record tool access, source set, configuration and reviewer disposition
- Stop and contain the workflow when protected information appears
- Apply disclosure, cybersecurity and legal review before operational use

Convenience never overrides statutory protection

Capability	Appropriate statistical role
Rules	Enforce explicit fields, classifications and thresholds
Machine learning	Estimate patterns when labelled evidence supports inference
Generative AI	Draft structured candidates, code, summaries and alternatives
Retrieval systems	Answer within an approved source collection with citations
Agentic workflows	Sequence bounded tools under permissions and logging
No AI	Use when evidence, risk or task structure does not justify assistance

Prompt engineering as specification

Prompt element	Required content
Purpose	Decision and intended product
Context	Statistical process, population and users
Evidence boundary	Approved sources and citation requirement
Output schema	Stable fields, formats and identifiers
Uncertainty rule	How to mark ambiguity and missing evidence
Prohibitions	Claims or actions the model must not perform
Acceptance criteria	Checks applied by independent reviewers
Stopping rule	Observable condition that ends or redirects work

What AI could do and should do

Work area	AI could do	AI should do under controls
Document extraction	Draft structured records	Use approved sources and schema-constrained output
Quality diagnostics	Suggest checks and anomalies	Generate candidate tests for independent review
Code development	Draft code and test cases	Operate in version control with human code review
Documentation	Draft summaries and alternatives	Cite verified findings in controlled templates
Workflow support	Sequence bounded tasks	Log actions and stop at authority boundaries

Technical capability alone does not justify use

Non-delegable specialist responsibilities

AI ASSISTED WORK

Statistician

- Define estimand, design and acceptance criteria
- Assess error, uncertainty and generalization
- Approve statistical fitness and release recommendation

Subject-matter specialist

- Confirm concepts and operational context
- Interpret substantive plausibility and consequences
- Resolve domain ambiguity and change mechanisms

AI provides candidates. Qualified people accept, correct or reject them.

Independent verification and correction

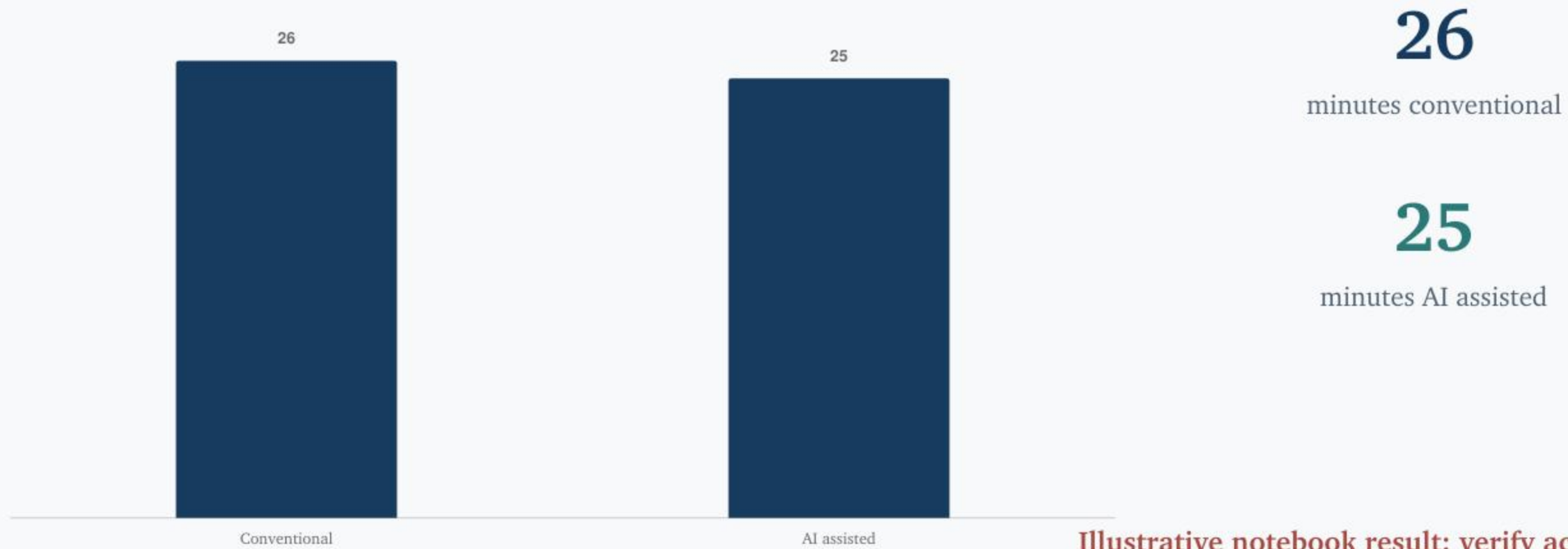
Stage	Reviewer action	Evidence produced
Preserve	Retain exact prompt, context and raw output	Immutable candidate record
Test structure	Check schema, identifiers and required fields	Automated validation results
Check evidence	Trace each material claim to approved sources	Source verification log
Apply expertise	Assess meaning, methods and consequences	Specialist review
Correct or reject	Record material errors and disposition	Correction log
Accept	Approve only the verified product	Named sign-off and limitations

Failure signals and stopping rules

Failure signal	Required response
Unsupported authority claim	Reject and check the governing instrument
Conflicting source interpretation	Preserve both readings and escalate
High extraction error rate	Stop batch processing and revise or abandon the method
Protected information detected	Stop, contain and follow incident procedure
Unreproducible output	Restrict use and return to a controlled workflow
No quality-adjusted gain	Use the conventional method

Quality adjusted efficiency

AI ASSURANCE



Illustrative notebook result: verify actual human effort

Practical 1 Actor authority map

Step	Learner action	Required evidence
Baseline	Complete ACT-01 through ACT-04 manually	Manual register and elapsed time
Prompt	Extract remaining actors within E01 to E10	Exact structured prompt
Verify	Check role and authority claims	Source checks and error log
Correct	Repair ACT-05 and ACT-06 candidate errors	Named correction owners
Accept	Produce the verified actor register	Final product and limitations

Practical 2 Asset and source validation

Asset	Primary risk	Specialist question
Annual crop survey	Coverage exclusions	Can excluded populations be estimated separately?
District spreadsheets	Inconsistent validation	Which rules and versions apply by district?
Satellite imagery	Systematic cloud missingness	How does missingness vary by crop and district?
Farmer registry	Incomplete identifiers	What are current omission, duplication and linkage errors?
Pilot results	Limited generalization	What supports transfer beyond tested domains?

Practical 3 Critical flow analysis

Flow	Critical failure	Fallback
District reports to ministry	Comparability failure	Quarantine and district query
Imagery to model features	Northern coverage bias	Flag unusable domains and use alternate evidence
Registry to linkage	False and missed matches	Stop linkage until thresholds pass
Integrated inputs to model	Unsupported transfer	Restrict the pilot domain
Reviewed estimates to release	Premature or unsafe release	Stop release and escalate

Practical 4 AI opportunity and pilot decision

Candidate task	Reference decision	Reason
Schema-constrained extraction	Controlled pilot	High value and independently verifiable
Draft validation tests	Controlled development use	Useful with review and execution tests
Determine legal authority	Prohibit autonomous use	Qualified interpretation required
Generate and release official estimates	Prohibit autonomous use	Method and release authority are non-delegable
Summarize verified findings	Controlled pilot	Acceptable when grounded, cited and reviewed

The Republic of Kintaba case

CASE

Operational need

- District crop estimates six weeks earlier
- Forty-eight districts with diverse conditions
- Users need uncertainty and revision flags

Evidence constraints

- Survey coverage exclusions
- Inconsistent district validation and registry identifiers
- Satellite pilot limited to twelve districts and two seasons

The case contains deliberate gaps and contradictions

Assessment and professional evidence

COMPLETION

Criterion	Weight	Minimum evidence
Scope and statistical purpose	15%	Defensible boundary and intended use
Actors assets and flows	20%	Complete source-traceable records
Quality risk and fitness	20%	Error mechanisms and consequences
AI allocation and prompting	15%	Bounded reproducible use
Verification and correction	20%	Independent checks and correction trail
Judgement and communication	10%	Recommendation, limits and named authority